Technical Note: Structural Dualism in Integer Architectures

From the Global Tree Root of the Canonical Triple-Graph to the Interwoven Lattice of the Goldbach Web

Enis Olgac
Independent Researcher, Böblingen, Germany
enis@digraphs.info | ORCID: 0009-0009-7560-6107

2026

Abstract

This technical note formalizes the structural dualism connecting two recently established discrete integer frameworks: the Canonical Triple-Graph (CTG) and the C-Matrix Cumulative Gap Lattice ($\mathcal{A} \times \mathcal{A}$). Historically, additive and operational problems over the integers—such as the Collatz Conjecture (1937) and the Goldbach Conjecture (1742)—have been treated as disjoint phenomena. We demonstrate that these two domains represent a perfect structural mirror. While the CTG organizes the positive integers into a deterministic, vertical branching tree engine where even integer trajectories are localized strangers dependent on isolated odd roots, the C-Matrix lattice flips the paradigm. By treating primes as multiplicative weavers, the cumulative gap mesh binds these independent branches into a highly organized, two-dimensional coordinate net. In both systems, moving from abstract number-theoretic complexity to deterministic fact relies entirely on finding an immutable geometric handle: the global tree root 1 in the CTG, and the canonical boundary anchor E on Row 0 of the Goldbach lattice.

Keywords. Canonical Triple-Graph, C-Matrix lattice, structural dualism, invariant handles, Goldbach conjecture, Collatz conjecture.

1 Introduction and Historical Context

Classical number theory routinely analyzes integers along a flat, one-dimensional number line. While computationally intuitive, this linear projection obscures the high-dimensional geometric frameworks that dictate operational behavior. The consequences of this paradigm limitation are visible in the historical intractability of two monumental open problems: the Goldbach Conjecture, originally posed in a 1742 letter to Leonhard Euler as an assertion on additive prime decompositions [1], and the Collatz Conjecture, popularized in 1937 by Lothar Collatz as an algorithmic mapping over arithmetic parity transitions [2].

Recent breakthroughs have bypassed these traditional bottlenecks by projecting the integer landscape onto rigid, high-dimensional coordinate networks. The mapping of the positive integers onto a deterministic directed graph establishes the Canonical Triple-Graph (CTG) architecture [3]. Conversely, translating the additive distribution of primes onto a static cumulative gap set maps the complete geometric landscape of the C-Matrix lattice [4].

This technical note establishes that these two architectures are not isolated systems; they represent a unified, dualistic structural mechanism. We formalize this duality by proving how the isolated branching pathways of the CTG are directly interwoven into a deterministic coordinate continuum within the Goldbach matrix mesh.

2 The Branching Engine vs. The Coordinate Net

The structural configuration of the integers undergoes a profound geometric inversion when moving from the operational chains of the CTG to the additive space of the C-Matrix lattice.

Definition 2.1 (The Localized Stranger Invariant)**.** Let $\mathcal{T}$ represent the global directed tree of the CTG [3]. For any distinct odd integer roots $o_1, o_2 \in \mathcal{O}$, let $\mathcal{B}(o_1)$ and $\mathcal{B}(o_2)$ define their respective downward vertical branches. Every even integer generated within $\mathcal{B}(o_1)$ at the base pillar of its odd root exists strictly inside an isolated arithmetic block. Within the localized boundaries of $\mathcal{T}$, these even integer trajectories share no direct operational edges:

$$\mathcal{B}(o_1) \cap \mathcal{B}(o_2) = \emptyset \quad \forall o_1 \neq o_2 \tag{1}$$

The evens of the graph are operational strangers, isolated on independent vertical rails.

Theorem 2.2 (The Lattice Inversion)**.** *Let $\mathcal{A} = \{p - 3 : p \in \mathcal{P}\}$ represent the cumulative prime gap set, and let $M_{\mathcal{A}} = \mathcal{A} \times \mathcal{A}$ define the complete C-Matrix lattice [4]. The odd primes act strictly as multiplicative weavers across this space. By forming the complete Cartesian product grid, the irregular intervals of the prime sequence collapse into a uniform, parity-quantized mesh. The previously isolated even branches $\mathcal{B}(o_i)$ are flattened and interwoven into a rigid two-dimensional continuum. Every even target $C = N - 6$ is transformed from an isolated node on a vertical branch into a hard geometric diagonal line that sweeps entirely across the network:*

$$L_C = \{(a, b) \in \mathcal{A} \times \mathcal{A} \mid a + b = C\} \tag{2}$$

Thus, the multiplicative irregularities of individual primes are structurally absorbed to build an absolute, interconnected additive web.

3 The Invariant Handles: In Theory and Practice

The translation of these systems from theoretical frameworks into proven facts depends entirely on identifying an irreplaceable geometric handle within each space.

Remark 3.1 (The Global Tree Root 1)**.** In the CTG architecture, the number 1 sits completely outside the periodic block structures [3]. It serves as the singular, absolute topological root of the directed graph. In hard algebraic practice, 1 acts as the universal initialization switchboard. Because every independent vertical block ultimately projects its operational trajectories down to this singular origin, the number 1 prevents the global network from fracturing into disjoint, unresolvable chains. It is the core invariant handle that makes the structural organization of the positive integers achievable.

Definition 3.2 (The Canonical Anchor E)**.** For any given even target C on the C-Matrix lattice, let E represent the canonical anchor defined as the smallest even integer strictly greater than the arithmetic midpoint $C/2$:

$$E = \begin{cases} \frac{C}{2} + 1 & \text{if } \frac{C}{2} \text{ is odd} \\ \frac{C}{2} + 2 & \text{if } \frac{C}{2} \text{ is even} \end{cases} \tag{3}$$

Theorem 3.3 (The Hard Physical Boundary Fact)**.** *The canonical anchor E is an irreplaceable structural handle pinned permanently to the matrix floor at Row 0 ($a = 0$). In concrete physical practice, it acts as a rigid retaining wall that isolates the problem:*

1. ***Parity Shielding:*** *Because the coordinate lattice steps exclusively by quantized even increments ($\Delta \geq 2$), the anchor E absorbs the alternation of $C/2$ internally, preventing column alignment drift.*
2. ***Active Zone Confinement:*** *Pinning E to Row 0 minimizes the vertical axis component, forcing the horizontal component to its absolute structural limit ($b_{\max} = d_{\max}(C)$). This instantly defines a closed, strictly finite active bounding rectangle $[a_{\min}, a_{\max}] \times [b_{\min}, b_{\max}]$.*

Proof. Because E locks the boundary walls of this finite active zone, the dual concurrent sweeps executed across the mesh are topologically trapped within a closed coordinate field. By simple planar separation, a monotone path tracing the lower sublevel envelope must intersect a monotone path tracing the upper superlevel invariant target diagonal $a + b = C$. Because the lattice possesses no fractional gaps or structural holes, the walks cannot bypass each other or fall through an unformed cell. The boundary anchor E forces a deterministic, physical collision at a valid intersection point $(a, b) \in \mathcal{A} \times \mathcal{A}$, closing the proof in practice. □

4 Conclusion

The structural dualism between the Canonical Triple-Graph and the Goldbach C-Matrix reveals a deep architectural symmetry within the integers. On its own, an even integer is an isolated arithmetic branch trajectory, locked inside a vertical block generated by its localized odd root. Yet, when projected across the cumulative gap lattice, these independent branches are interwoven by the primes into a deterministic, two-dimensional spatial net.

Whether tracking down vertical graph transitions to the absolute root 1, or trapping horizontal lattice sweeps against the rigid boundary anchor E on Row 0, the infinite complexity of numbers is shown to be governed by hard, invariant geometric handles.

References

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